



DSA0602-DATA HANDLING AND DATA VISUALIZATION

SLOT A

Dr. ALMAS BEGUM

PRESENTED BY:

SURU SAI KEERTHI [192424368]

SET-13

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

```
library(ggplot2)
```

```
library(maps)
```

```
city_data <- data.frame(
```

```
  City = c("City A", "City B", "City C"),
```

```
  Population = c(500000, 700000, 600000),
```

```
  Avg_Temperature = c(75, 68, 80),
```

```
  Elevation = c(1000, 800, 1200),
```

```
  Longitude = c(77.6, 78.4, 79.2),
```

```
  Latitude = c(12.9, 13.1, 13.5)
```

```
)
```

```
world_map <- map_data("world")
```

```
ggplot() +
```

```
  geom_map(
```

```
    data = world_map,
```

```
    map = world_map,
```

```
    aes(map_id = region),
```

```
    fill = "lightgray",
```

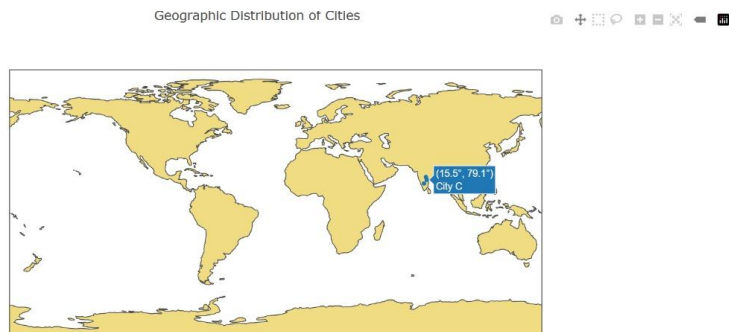
```
    color = "white"
```

```
) +
```

```

geom_point(
  data = city_data,
  aes(x = Longitude, y = Latitude, size = Population, color = City),
  alpha = 0.8
) +
geom_text(
  data = city_data,
  aes(x = Longitude, y = Latitude, label = City),
  vjust = -1
) +
coord_quickmap() +
labs(
  title = "Geographic Distribution of Cities",
  x = "Longitude",
  y = "Latitude"
) +
theme_minimal()

```



2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```

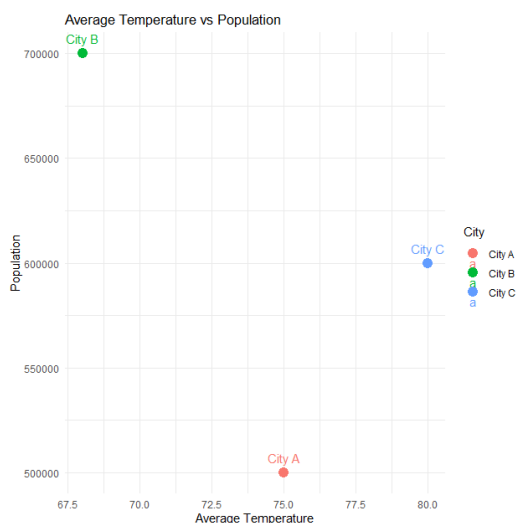
library(ggplot2)
city_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80)
)

```

```

ggplot(city_data,
  aes(x = Avg_Temperature,
    y = Population,
    color = City)) +
  geom_point(size = 4) +
  geom_text(aes(label = City), vjust = -1) +
  labs(
    title = "Average Temperature vs Population",
    x = "Average Temperature",
    y = "Population"
  ) +
  theme_minimal()
cat("Insight:\n")
cat("The scatter plot shows the relationship between average temperature and population.\n")
cat("No strong correlation is observed because the dataset contains only three cities.\n")

```



3. Build a table to display the geographic data for each city. Label the table elements.

```

city_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),

```

```

Elevation = c(1000, 800, 1200)
)
names(city_data) <- c(
  "City",
  "Population",
  "Average Temperature",
  "Elevation"
)
print("Geographic Data")
print(city_data)

```

	City	Population	Average Temperature	Elevation
1	City A	5e+05	75	1000
2	City B	7e+05	68	800
3	City C	6e+05	80	1200

4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

```

library(shiny)
library(ggplot2)
city_data <- data.frame(
  City = c("City A", "City B", "City C"),
  Population = c(500000, 700000, 600000),
  Avg_Temperature = c(75, 68, 80),
  Elevation = c(1000, 800, 1200),
  Longitude = c(77.6, 78.4, 79.2),
  Latitude = c(12.9, 13.1, 13.5)
)
ui <- fluidPage(
  titlePanel("Geographic Data Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "city",
        "Select City:",

```

```

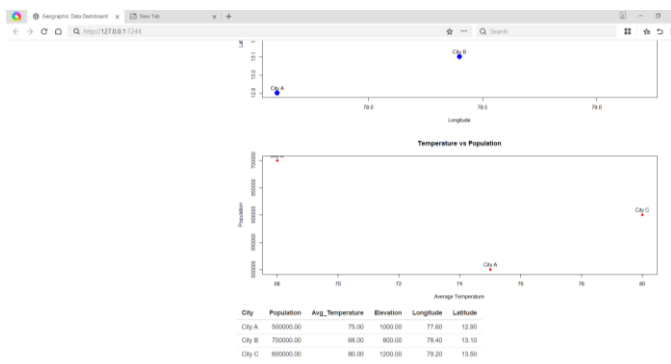
    choices = c("All", city_data$City)
  )
), mainPanel(
  plotOutput("mapChart"),
  plotOutput("scatterPlot"),
  tableOutput("cityTable")
)
)
)
server <- function(input, output){
  filtered_data <- reactive({
    if(input$city=="All"){
      city_data
    } else{
      city_data[city_data$City==input$city,]
    }
  })
  output$mapChart <- renderPlot({
    plot(
      filtered_data()$Longitude,
      filtered_data()$Latitude,
      pch=19,
      cex=2,
      col="blue",
      xlab="Longitude",
      ylab="Latitude",
      main="City Locations"
    )
    text(filtered_data()$Longitude,
          filtered_data()$Latitude,
          labels=filtered_data()$City,

```

```

      pos=3)
    })
    output$scatterPlot <- renderPlot({
      plot(
        filtered_data()$Avg_Temperature,
        filtered_data()$Population,
        pch=19,
        col="red",
        xlab="Average Temperature",
        ylab="Population",
        main="Temperature vs Population"
      )
      text(filtered_data()$Avg_Temperature,
            filtered_data()$Population,
            labels=filtered_data()$City,
            pos=3)
    })
    output$cityTable <- renderTable({
      filtered_data()
    })
  }
  shinyApp(ui, server)

```



SET-14

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart

elements.

```
library(ggplot2)
```

```
survey_data <- data.frame(
```

```
  Respondent = c("R1","R2","R3","R4","R5"),
```

```
  Q1 = c("Yes","No","Yes","Yes","No"),
```

```
  Q2 = c("Good","Average","Good","Excellent","Average"),
```

```
  Q3 = c("Satisfied","Neutral","Satisfied","Satisfied","Neutral")
```

```
)
```

```
q1_count <- as.data.frame(table(survey_data$Q1))
```

```
colnames(q1_count) <- c("Answer","Count")
```

```
ggplot(q1_count,
```

```
  aes(x = "Question 1",
```

```
    y = Count,
```

```
    fill = Answer)) +
```

```
geom_bar(stat = "identity") +
```

```
labs(
```

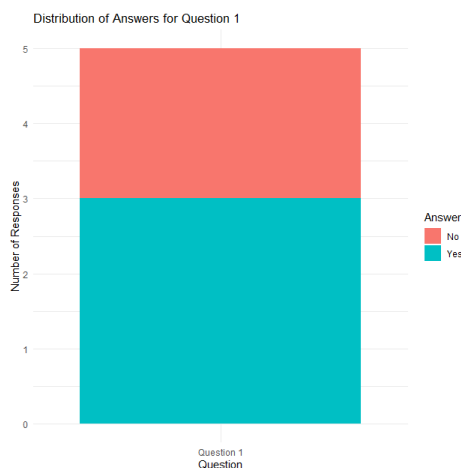
```
  title = "Distribution of Answers for Question 1",
```

```
  x = "Question",
```

```
  y = "Number of Responses"
```

```
) +
```

```
theme_minimal()
```



2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

```

library(fmsb)

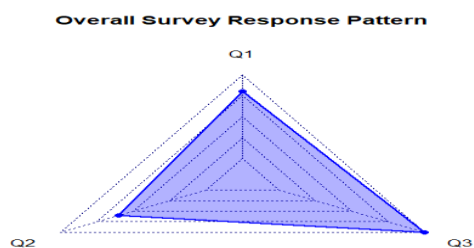
survey_scores <- data.frame(
  Q1 = c(4),
  Q2 = c(3),
  Q3 = c(5)
)

radar <- rbind(
  c(5,5,5),
  c(0,0,0),
  survey_scores
)

colnames(radar) <- c("Q1","Q2","Q3")

radarchart(
  radar,
  pcol = "blue",
  pfcol = rgb(0,0,1,0.3),
  plwd = 2,
  title = "Overall Survey Response Pattern"
)

```



3. Build a table to show the survey response data for each respondent. Label the table elements.

```

survey_data <- data.frame(
  Respondent = c("R1","R2","R3","R4","R5"),
  Q1 = c("Yes","No","Yes","Yes","No"),
  Q2 = c("Good","Average","Good","Excellent","Average"),
  Q3 = c("Satisfied","Neutral","Satisfied","Satisfied","Neutral")
)

```



```
)
names(survey_data) <- c(
  "Respondent",
  "Question 1",
  "Question 2",
  "Question 3"
)
print("Survey Response Data")
print(survey_data)
```

	Respondent	Question 1	Question 2	Question 3
1	R1	Yes	Good	Satisfied
2	R2	No	Average	Neutral
3	R3	Yes	Good	Satisfied
4	R4	Yes	Excellent	Satisfied
5	R5	No	Average	Neutral

4. Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

```
library(shiny)
library(ggplot2)
library(fmsb)

survey_data <- data.frame(
  Respondent = c("R1", "R2", "R3", "R4", "R5"),
  Q1 = c("Yes", "No", "Yes", "Yes", "No"),
  Q2 = c("Good", "Average", "Good", "Excellent", "Average"),
  Q3 = c("Satisfied", "Neutral", "Satisfied", "Satisfied", "Neutral")
)

ui <- fluidPage(
  titlePanel("Survey Response Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "question",
        "Select Question",
        choices = c("Q1", "Q2", "Q3")
      )
    )
  )
)
```

```

    )
  ),mainPanel(
    plotOutput("stackedBar"),
    plotOutput("radar"),
    tableOutput("surveyTable")
  )))
server <- function(input, output){
  output$stackedBar <- renderPlot({
    counts <- as.data.frame(table(survey_data[[input$question]]))
    colnames(counts) <- c("Answer","Count")
    ggplot(counts,
      aes(x=input$question,
          y=Count,
          fill=Answer))+
    geom_bar(stat="identity")+
    labs(
      title=paste("Responses for",input$question),
      x="Question",
      y="Count"
    )+
    theme_minimal()
  })
  output$radar <- renderPlot({
    radar <- rbind(
      c(5,5,5),
      c(0,0,0),
      c(4,3,5)
    )
    colnames(radar) <- c("Q1","Q2","Q3")
    radarchart(
      radar,

```

```

    pcol="blue",
    pfc=rgb(0,0,1,0.3),
    plwd=2
  )
})
output$surveyTable <- renderTable({
  survey_data
})
}

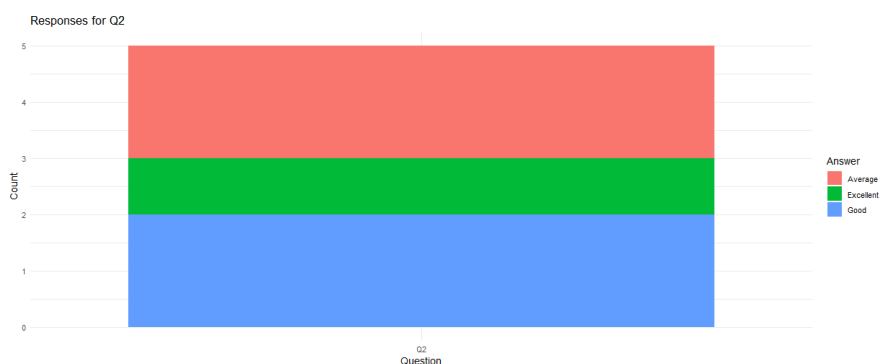
```

```
shinyApp(ui, server)
```

Survey Response Dashboard

Select Question

Q2



Respondent	Q1	Q2	Q3
R1	Yes	Good	Satisfied
R2	No	Average	Neutral
R3	Yes	Good	Satisfied
R4	Yes	Excellent	Satisfied
R5	No	Average	Neutral

SET-15

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

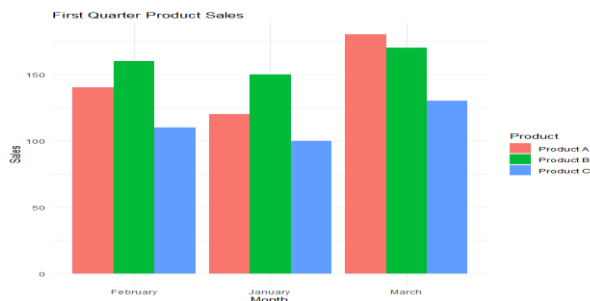
```
library(ggplot2)
```

```
sales_data <- data.frame(
```

```

Month = rep(c("January", "February", "March"), each = 3),
Product = rep(c("Product A", "Product B", "Product C"), 3),
Sales = c(120, 150, 100, 140, 160, 110, 180, 170, 130)
)
ggplot(sales_data,
  aes(x = Month,
    y = Sales,
    fill = Product)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "First Quarter Product Sales",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()

```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```

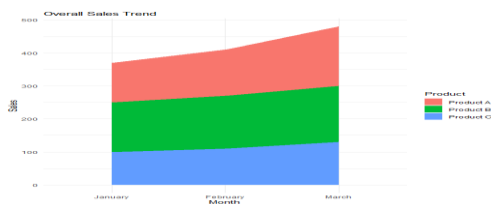
library(ggplot2)
sales_data <- data.frame(
  Month = factor(c("January", "February", "March"),
    levels = c("January", "February", "March")),
  ProductA = c(120, 140, 180),
  ProductB = c(150, 160, 170),
  ProductC = c(100, 110, 130)
)
stack_data <- data.frame(

```

```

Month = rep(sales_data$Month, 3),
Product = rep(c("Product A","Product B","Product C"), each = 3),
Sales = c(
  sales_data$ProductA,
  sales_data$ProductB,
  sales_data$ProductC
))ggplot(stack_data,
  aes(x = Month,
    y = Sales,
    fill = Product,
    group = Product)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()

```



3. Build a table to show the monthly sales data for each product. Label the table elements.

```

sales_data <- data.frame(
  Month = c("January","February","March"),
  Product_A = c(120,140,180),
  Product_B = c(150,160,170),
  Product_C = c(100,110,130)
)
names(sales_data) <- c(
  "Month",
  "Product A"

```

```

"Product B",
"Product C"
)
print("Monthly Sales Data")
print(sales_data)

```

```

      Month Product A Product B Product C
1 January      120      150      100
2 February     140      160      110
3 March        180      170      130

```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```

library(shiny)
library(ggplot2)
sales_data <- data.frame(
  Month = rep(c("January", "February", "March"), each = 3),
  Product = rep(c("Product A", "Product B", "Product C"), 3),
  Sales = c(120, 150, 100, 140, 160, 110, 180, 170, 130)
)
ui <- fluidPage(
  titlePanel("Sales Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "product",
        "Select Product",
        choices = c("All", unique(sales_data$Product))
      )
    ),
    mainPanel(
      plotOutput("barChart"),
      plotOutput("areaChart"),
      tableOutput("salesTable")
    )
  )
)

```

```

)))server <- function(input, output){
filtered <- reactive({
  if(input$product == "All")
    sales_data
  else
    subset(sales_data, Product == input$product)
})
output$barChart <- renderPlot({
  ggplot(filtered(),
    aes(x = Month,
        y = Sales,
        fill = Product)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Grouped Bar Chart",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
})
output$areaChart <- renderPlot({
  ggplot(filtered(),
    aes(x = Month,
        y = Sales,
        fill = Product,
        group = Product)) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Stacked Area Chart",
    x = "Month",
    y = "Sales"
  )
})

```

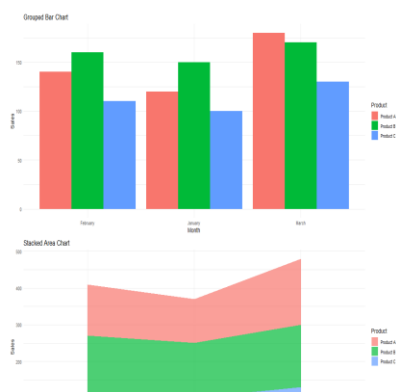
```

    ) +
    theme_minimal()
  })output$salesTable <- renderTable({
    filtered()
  })
}
shinyApp(ui, server)

```

Sales Dashboard

Select Product
All



Month	Product	Sales
January	Product A	120.00
January	Product B	150.00
January	Product C	100.00
February	Product A	140.00
February	Product B	160.00
February	Product C	110.00
March	Product A	180.00
March	Product B	170.00
March	Product C	130.00

SET-16

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

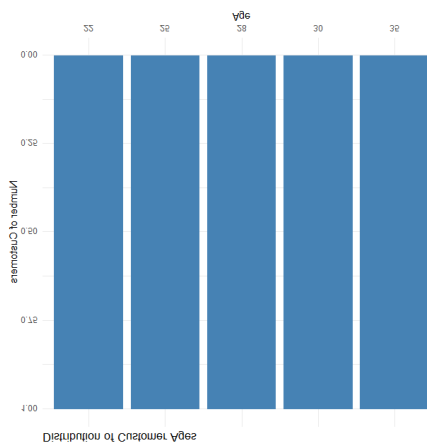
```

library(ggplot2)

customer_data <- data.frame(
  Customer = c("C1","C2","C3","C4","C5"),
  Age = c(22,25,30,28,35),
  Gender = c("Male","Female","Female","Male","Female")
)

ggplot(customer_data,
  aes(x = factor(Age))) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal()

```

2. Generate a pie chart to represent the distribution of customers by gender.

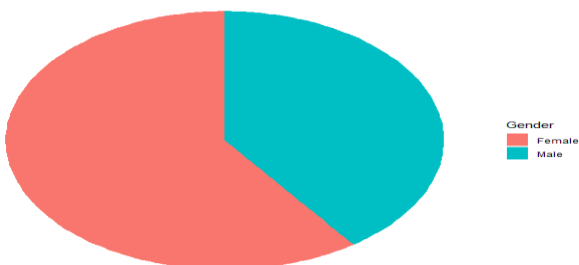
```
library(ggplot2)

customer_data <- data.frame(
  Customer = c("C1", "C2", "C3", "C4", "C5"),
  Age = c(22, 25, 30, 28, 35),
  Gender = c("Male", "Female", "Female", "Male", "Female")
)

gender_count <- as.data.frame(table(customer_data$Gender))
colnames(gender_count) <- c("Gender", "Count")

ggplot(gender_count,
  aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(
    title = "Distribution of Customers by Gender"
  ) +
  theme_void()
```

Distribution of Customers by Gender



3. Build a table to show the demographic information for each customer. Label the table elements.

```
customer_data <- data.frame(  
  Customer = c("C1","C2","C3","C4","C5"),  
  Age = c(22,25,30,28,35),  
  Gender = c("Male","Female","Female","Male","Female")  
)  
names(customer_data) <- c(  
  "Customer ID",  
  "Age",  
  "Gender"  
)  
print("Customer Demographic Information")  
print(customer_data)
```

	Customer ID	Age	Gender
1	C1	22	Male
2	C2	25	Female
3	C3	30	Female
4	C4	28	Male
5	C5	35	Female

4. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

```
library(shiny)  
library(ggplot2)  
customer_data <- data.frame(  
  Customer = c("C1","C2","C3","C4","C5"),  
  Age = c(22,25,30,28,35),  
  Gender = c("Male","Female","Female","Male","Female")  
)  
ui <- fluidPage(  
  titlePanel("Customer Demographics Dashboard"),  
  sidebarLayout(  
    sidebarPanel(  
      selectInput(  
        "gender",
```

```

    "Select Gender",
    choices = c("All", unique(customer_data$Gender))
  )
),
mainPanel(
  plotOutput("barChart"),
  plotOutput("pieChart"),
  tableOutput("customerTable")
)
)
)
server <- function(input, output){
  filtered <- reactive({
    if(input$gender == "All")
      customer_data
    else
      subset(customer_data, Gender == input$gender)
  })
  output$barChart <- renderPlot({
    ggplot(filtered(),
      aes(x = factor(Age))) +
    geom_bar(fill = "steelblue") +
    labs(
      title = "Customer Age Distribution",
      x = "Age",
      y = "Count"
    ) +
    theme_minimal()
  })
  output$pieChart <- renderPlot({
    gender_count <- as.data.frame(table(filtered())$Gender))

```

```

colnames(gender_count) <- c("Gender","Count")

ggplot(gender_count,
       aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  theme_void() +
  labs(title = "Gender Distribution")
})

output$customerTable <- renderTable({
  filtered()
})
}

shinyApp(ui, server)

```

Customer Demographics Dashboard

Select Gender
All



Customer	Age	Gender
C1	22.00	Male
C2	25.00	Female
C3	30.00	Female
C4	28.00	Male
C5	35.00	Female

SET-17

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

```

library(ggplot2)

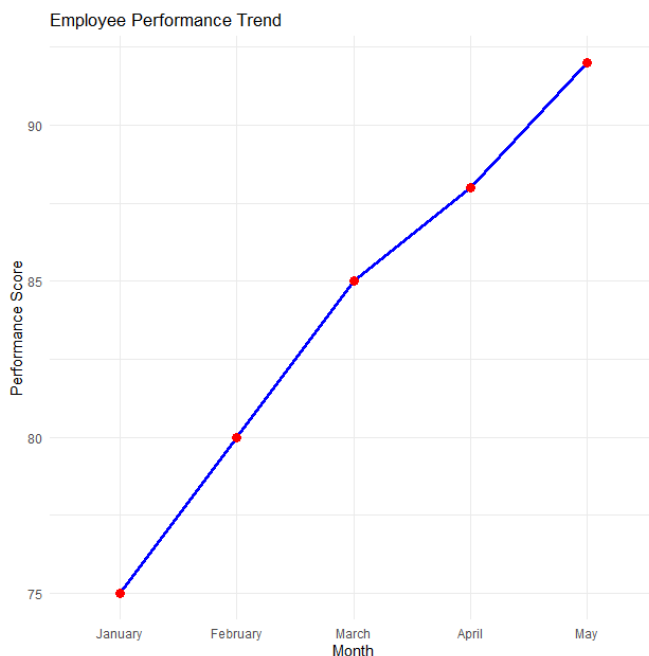
employee_data <- data.frame(
  Month = factor(c("January","February","March","April","May"),
    levels = c("January","February","March","April","May")),

```

```

Performance = c(75,80,85,88,92)
)
ggplot(employee_data,
  aes(x = Month,
      y = Performance,
      group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(size = 3, color = "red") +
  labs(
    title = "Employee Performance Trend",
    x = "Month",
    y = "Performance Score"
  ) +
  theme_minimal()

```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

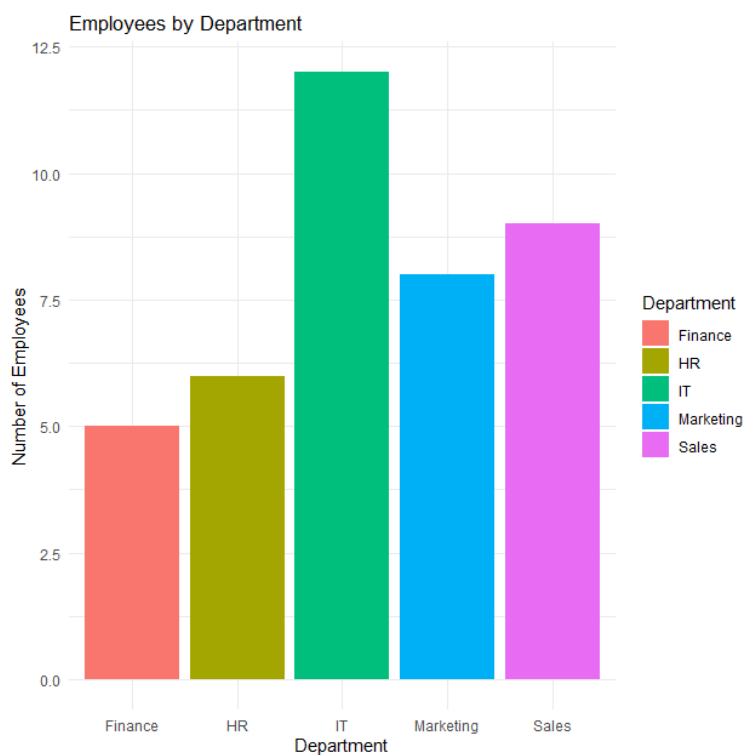
```

library(ggplot2)
department_data <- data.frame(
  Department = c("HR", "IT", "Finance", "Sales", "Marketing"),
  Employees = c(6, 12, 5, 9, 8)
)

```

)

```
ggplot(department_data,  
  aes(x = Department,  
    y = Employees,  
    fill = Department)) +  
geom_bar(stat = "identity") +  
labs(  
  title = "Employees by Department",  
  x = "Department",  
  y = "Number of Employees"  
) +  
theme_minimal()
```



3. Build a table to display the performance data for each employee. Label the table elements.

```
employee_data <- data.frame(  
  Employee_ID = c("E101","E102","E103","E104","E105"),  
  Employee_Name = c("John","Alice","David","Emma","James"),  
  Department = c("HR","IT","Finance","Sales","Marketing"),  
  Performance_Score = c(75,80,85,88,92)
```

)

```
names(employee_data) <- c(
```

```
  "Employee ID",
```

```
  "Employee Name",
```

```
  "Department",
```

```
  "Performance Score"
```

)

```
print("Employee Performance Data")
```

```
print(employee_data)
```

	Employee ID	Employee Name	Department	Performance Score
1	E101	John	HR	75
2	E102	Alice	IT	80
3	E103	David	Finance	85
4	E104	Emma	Sales	88
5	E105	James	Marketing	92

4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

```
library(shiny)
```

```
library(ggplot2)
```

```
employee_data <- data.frame(
```

```
  Employee_ID = c("E101","E102","E103","E104","E105"),
```

```
  Employee_Name = c("John","Alice","David","Emma","James"),
```

```
  Department = c("HR","IT","Finance","Sales","Marketing"),
```

```
  Performance_Score = c(75,80,85,88,92)
```

)

```
trend_data <- data.frame(
```

```
  Month = factor(c("January","February","March","April","May"),
```

```
    levels=c("January","February","March","April","May")),
```

```
  Performance = c(75,80,85,88,92)
```

)

```
ui <- fluidPage(
```

```
  titlePanel("Employee Performance Dashboard"),
```

```
  sidebarLayout(
```

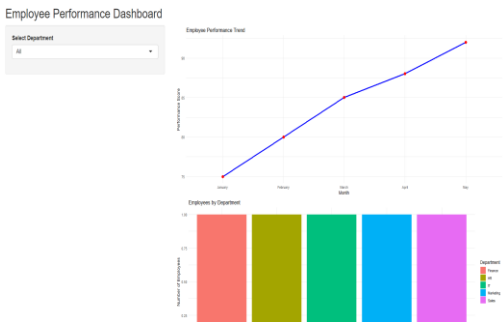
```

sidebarPanel(
  selectInput(
    "dept",
    "Select Department",
    choices = c("All", unique(employee_data$Department))),
  mainPanel(
    plotOutput("lineChart"),
    plotOutput("barChart"),
    tableOutput("empTable")
  )))server <- function(input, output){
  filtered <- reactive({
    if(input$dept=="All")
      employee_data
    else
      subset(employee_data, Department==input$dept)
  })
  output$lineChart <- renderPlot({
    ggplot(trend_data,
      aes(Month, Performance, group=1))+
    geom_line(color="blue", linewidth=1.2)+
    geom_point(color="red", size=3)+
    labs(
      title="Employee Performance Trend",
      x="Month",
      y="Performance Score"
    )+
    theme_minimal()
  })
  output$barChart <- renderPlot({
    dept_count <- as.data.frame(table(filtered())$Department))
    colnames(dept_count) <- c("Department", "Employees")
  })
}

```



```
ggplot(dept_count,
      aes(Department,
          Employees,
          fill=Department))+
geom_bar(stat="identity")+
labs(
  title="Employees by Department",
  x="Department",
  y="Number of Employees"
)+
theme_minimal()
})
output$empTable <- renderTable({
  filtered()
})
}
shinyApp(ui, server)
```



Department			
Employee_ID	Employee_Name	Department	Performance_Score
E101	John	HR	75.00
E102	Alice	IT	80.00
E103	David	Finance	85.00
E104	Emma	Sales	88.00
E105	James	Marketing	92.00